Scientific Article Secion Classification

August 2025

1 Literature Review

1.1 Related Work

Research on scientific article section classification has moved forward step by step from 2018 to 2024.

In 2018, (Jin and Szolovits, 2018) introduced HSLN (hierarchical BiLSTM+CRF), reaching about **92.6** F1 on PubMed-20k. In 2019, (Cohan et al., 2019) used SciBERT with contextual encoding and set a stronger baseline on PubMed-20k at **92.9** F1 (and **83.1** F1 on CSAbstruct). In 2020, (Yamada et al., 2020) moved to span-level decoding with a neural semi-Markov CRF, lifting PubMed-20k to **93.1** F1 (sentence) and **84.3** F1 (span). In 2021, (Shang et al., 2021) added dynamic local attention over spans, pushing NICTA to **86.8** F1 (sentence) and **62.9** F1 (span).

To go beyond single datasets, 2022 saw (Brack et al., 2022) apply cross-domain multi-task learning (SciBERT-HSLN), with large gains on full-paper corpora (e.g., Dr. Inventor **83.4** F1, ART/CoreSC **58.8** accuracy). In 2023, (Tokala et al., 2023) introduced HIHT, a label-informed hierarchical Transformer, improving PubMed-20k to about **93.2** F1; in the same year, (Hu et al., 2023) showed prompt-based BERT can reach **95.1** F1 on PubMed-200k and stay above **90** F1 in cross-domain tests. Most recently, 2024 reframed the task with LLM-SSC (Lan et al., 2024) (multi-label, sequential), achieving **92.5** F1 on PubMed-20k and **76.8** F1 on CS abstracts.

Overall, the field moved from hierarchical BiLSTMs to domain-pretrained BERT, then to span/label-aware designs, multi-task transfer, and now prompt/LLM methods—rising from roughly **92** F1 (2018) to **95+** F1 (2023) while shifting focus toward robustness and generalization.

Also checking in the detailed improvements for each SOTA papers in Table 1.1.

1.2 Datasets

The table 1.2 lists five data levels. Abstract-level: one abstract is a sample and we tag each sentence; split by abstract. Sentence-level: standalone sentences; split by their source document. Section-level: one section (e.g., Introduction) is a sample and the label covers the whole section; split by paper. Paper-level: a full paper with sentence tags; split by paper. Proposal-level: one proposal/idea with sentence tags; split by proposal.

Domain-specific rows (biomed, CS, NLP) check in-domain performance; cross-domain rows (CoreSC/ART[2] (Liakata et al., 2010), Emerald 110k (Stead et al., 2019), arXiv subsets (Cohan et al., 2018)) test transfer. For scale, Academic Section Classification, PubMed 200k RCT (Dernoncourt and Lee, 2017) and arXiv (Cohan et al., 2018) are largest; newer sets include BioRC800 (Lan et al., 2024), SPACE-IDEAS (García-Silva et al., 2024) and Academic Section Classification Datasets (Hoepner, 2025).

For ease of access and scale, we use the **Academic Section Classification dataset** on Hugging Face (one-line load, fixed splits). For very large pretraining, arXiv subsets are an option, though section labels must be mapped from headings. PubMed 200k RCT also has official splits and strong baselines. In this work, we proceed with Academic Section Classification Datasets.

[1]OS = observational study abstracts; RCT = randomized controlled trial abstracts. OS→RCT: train on OS, test on RCT. RCT→OS: train on RCT, test on OS.
[2]RCT = Randomized Controlled Trial

Year	Paper (method)	Dataset & Index	Results (Paper Link)
2018	(Jin and Szolovits, 2018)(EMNLP), *Hierarchical Neural Networks for Sequential Sentence Classification in Medical Scientific Abstracts* (hierarchical Bi-LSTM encoder + CRF)	PubMed-20k RCT (weighted-F1), PubMed-200k RCT (weighted-F1), NICTA-PIBOSO (weighted-F1)	**92.6** (20k) / **93.9** (200k), **84.7** (NICTA). (ACL Anthology)
2019	(Cohan et al., 2019)(EMNLP), *Pretrained Language Models for Sequential Sentence Classification* (SciBERT joint-sentence encoding + CRF)	PubMed-20k RCT (micro-F1), CSAbstruct (micro-F1), NICTA-PIBOSO (micro-F1)	**92.9** / **83.1** / **84.8**. (ACL Anthology)
2020	(Yamada et al., 2020) (Findings of EMNLP), *Sequential Span Classification with Neural Semi-Markov CRFs for Biomedical Abstracts* (BERT + semi-Markov CRF, span-level decoding)	PubMed-20k (Sentence-F1 / Span-F1), NICTA-PIBOSO (Sentence-F1 / Span-F1)	**93.1** / **84.3** (PubMed), **84.4** / **58.7** (NICTA). (ACL Anthology)
2021	(Shang et al., 2021) (ACL), *A Span-based Dynamic Local Attention Model for Sequential Sentence Classification* (Bi-GRU + span-aware attention + CRF)	PubMed-20k (Sentence-F1 / Span-F1), NICTA-PIBOSO (Sentence-F1 / Span-F1)	**92.8** / **84.5** (PubMed), **86.8** / **62.9** (NICTA). (ACL Anthology)
2022	(Brack et al., 2022)(JCDL), *Cross-Domain Multi-Task Learning for Sequential Sentence Classification in Research Papers* (Multi-task SciBERT-HSLN (MULT-GRP))	PubMed-20k (weighted-F1), NICTA (weighted-F1), Dr. Inventor (weighted-F1), ART/CoreSC (Accuracy)	Unified single-task: **92.9** / **84.9** / **78.0** / **58.0**; Best multi-task (MULT-GRP): **93.0** / **86.1** / **83.4** / **58.8**. (JCDL)
2023	(Tokala et al., 2023) (Expert Systems), *Label-Informed Hierarchical Transformers for Sequential Sentence Classification in Scientific Abstracts* (Label-informed hierarchical Transformer + CRF)	PubMed-20k RCT (macro-F1), NICTA-PIBOSO (macro-F1), CS-Abstruct (macro-F1)	**93.19** (PubMed-20k), **86.77** (NICTA), **83.29** (CS-Abstruct). (Expert Systems)
	(Hu et al., 2023) (Bioinformatics) *Towards More Generalizable and Accurate Sentence Classification in Medical Abstracts with Less Data* (Prompt-learning with BERT variants, few-shot)	PubMed-20k RCT (F1), PubMed-200k RCT (F1), OS dataset (cross-domain)	**94.01** (20k) / **95.08** (200k); 20% data: 92.66 / 92.63; Cross-domain: OS→RCT **92.03**, RCT→OS **90.65**. (Bioinformatics)
2024	(Lan et al., 2024) (Findings of EMNLP), *Multi-label Sequential Sentence Classification via Large Language Model* (LLM-SSC: ICL + PEFT/LoRA + contrastive loss)	PubMed-20k RCT (micro/macro-F1), CS-Abstruct (micro/macro-F1), ART-CoreSC (micro/macro-F1)	**92.5** / **87.9** (PubMed-20k), **76.8** / **71.6** (CS-Abstruct), **52.4** / **28.2** (ART). (ACL Anthology)

Table 1.1: Key papers in scientific article section classification (2018–2024). Bold numbers are reported best results; links are clickable.[1]

Table 1.2: Datasets used for scientific article section classification

Category	Dataset	Size (unit)	Labels	Description	Introduced by / Source	Used in Papers
Biomedical	PubMed 20k / 200k RCT	20k / 200k abs	5: Background Objective Method Result Conclusion	RCT abstracts; standard biomedical benchmark	(Dernoncourt and Lee, 2017)	HSLN (2018) SciBERT (2019) Span-SemiCRF (2020) Span-Attention (2021) SciBERT-HSLN (2022) LIHT (2023) Prompt-BERT (2023) LLM-SSC (2024)
	NICTA-PIBOSO	1k abs	6: Population Intervention Background Outcome Study design Other	Evidence-based medicine abstracts with PIBOSO labels (multi-label)	(Kim et al., 2010)	HSLN (2018) SciBERT (2019) Span-SemiCRF (2020) LIHT (2023)
	Observational Abstracts	20k abs	5: Background Objective Method Result Conclusion	Observational-study abstracts; same scheme as RCT	(Hu et al., 2023)	Prompt-BERT (2023)
	BioRC800	800 abs	6 (multi-label): Background Objective Methods Results Conclusions Other	Biomedical abstracts with multi-label sentence roles	(Lan et al., 2024)	LLM-SSC (2024)
Domain-specific	CSAbstruct	2,189 abs	5: Background Objective Method Result Other	Computer Science abstracts; RCT-style roles	(Cohan et al., 2019)	SciBERT (2019) SciBERT-HSLN (2022) LIHT (2023) LLM-SSC (2024)
	Dr.Inventor	∼40 papers	5: Background Challenge Approach Outcome Future Work	Computer Graphics full papers with discourse labels	(Ronzano and Saggion, 2015)	SciBERT-HSLN (2022)
	AAN (ACL Anthology Network)	∼11k papers	Varies (derived)	NLP papers; coarse roles often derived from section titles	(Bird et al., 2008) (Radev et al., 2013)	Derived section/discourse studies
	SPACE-IDEAS	176 / 1,020 ideas(=proposals)	5: Challenge Proposal Elaboration Benefits Context	Aerospace proposals; gold and larger LLM-assisted sets	(García-Silva et al., 2024)	Released with baselines

Category	Dataset	Size (unit)	Labels	Description	Introduced by / Source	Used in Papers
	Academic Section Classification Dataset	~131k section	5: Introduction Background Methodology Experiments & Results Conclusion	ACL-OCL based section dataset (Hugging Face)	(Hoepner, 2025)	New SSC benchmark (NLP)
Cross-domain	CoreSC/ ART	225 papers	11: Hypothesis Motivation Goal Object Background Method Experiment Model Observation Result Conclusion	Full papers across sciences; fine-grained discourse roles	(Liakata et al., 2010)	Span-Attention (2021) SciBERT-HSLN (2022)
	Emerald 110k	~103k abs	7: Purpose Design/method/approach Findings Originality/value Social implications Practical implications Research limitations/implications	Multidisciplinary structured abstracts	(Stead et al., 2019)	SciBERT-HSLN (2022)
	arXiv(subsets)	~215k papers	5 (mapped): Introduction Related Work Methods Results Conclusion	Preprints across many fields; labels mapped from headings	(Cohan et al., 2018)	LLM-SSC (2024) summarization + sectioning

Note: abs = abstract, sents = sentences, sections = one labeled section from a paper (e.g., Introduction), ideas = proposal submissions

Note.

- *Abstract-level datasets:* PubMed 20k/200k RCT, NICTA-PIBOSO, Observational Abstracts, BioRC800, CSAbstract, Emerald 110k.
- *Sentence-level datasets:* CoreSC/ART.
- *Paper-level datasets:* Dr.Inventor, AAN(ACL Anthology Network), arXiv subsets.
- *Section-level datasets:* Academic Section Classification Datasets.
- *Proposal-level dataset:* SPACE-IDEAS.s

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